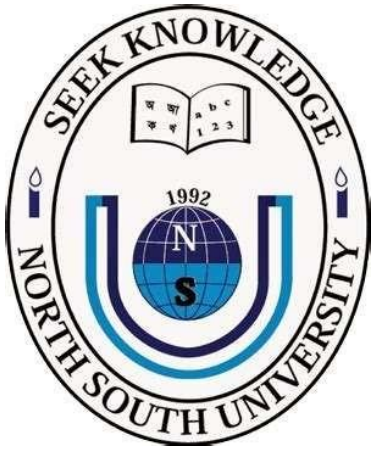
Analog Electronics II: Op-amp & 555 Timer



Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 14

OSCILLATORS

Reference:

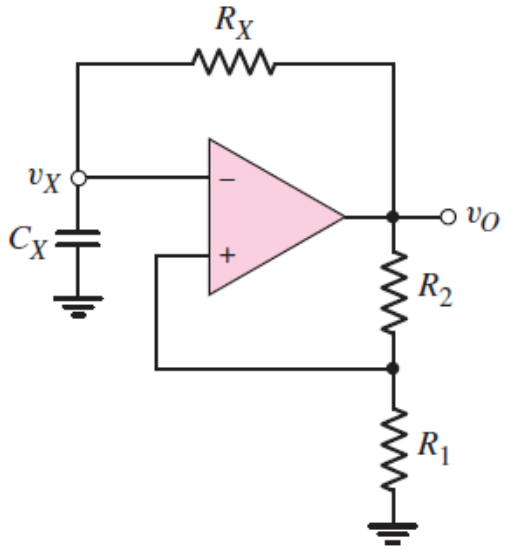
1. Lecture Slides
2. Microelectronic Circuit Analysis and Design, by Donald Neaman, Chapter 15 (15.2)
3. Operational Amplifiers and Linear Integrated Circuits, by Robert Coughlin Chapter Chapter 6 (6.1-6.4)

NONSINUSOIDAL OSCILLATORS

Chapter 15.4. from Microelectronic Circuit Analysis and Design by Donald A Neaman 4th Ed.

Many applications, especially digital electronic systems, use a nonsinusoidal squarewave oscillator to provide a clock signal for the system.
In other applications, a single pulse of known height and width is used to initiate a particular set of functions.

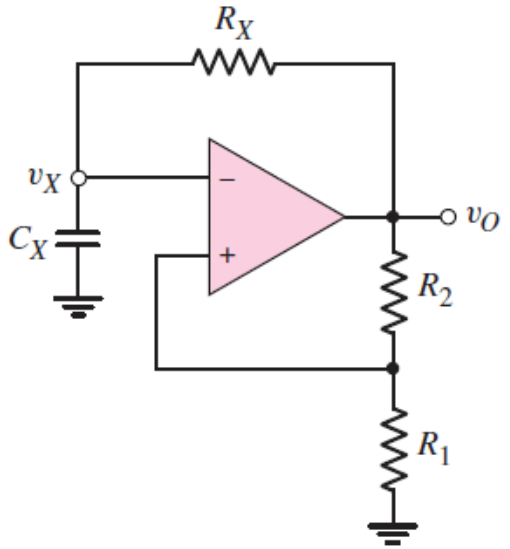
Schmitt Trigger Oscillator



The Schmitt trigger can be used in an oscillator circuit to generate a square-wave output signal.

Schmitt Trigger Oscillator

This circuit has no stable states. It is therefore called an **astable multivibrator.**



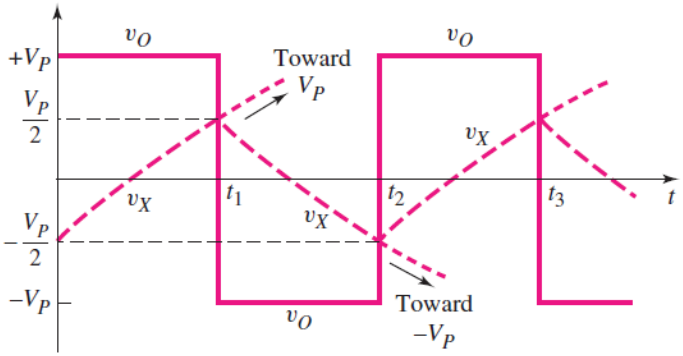
The high saturated output denoted by $V_H = V_P$
The low saturated output denoted by $V_L = -V_P$

$$R_1 = R_2 \equiv R$$

If v_O is low, or $v_O = -V_P$, then $v+ = -(1/2)V_P$.

When v_X drops just slightly below v+, the output switches high so that $v_O = +V_P$ and $v+ = +(1/2)V_P$

The R_XC_X network sees a positive step-increase in voltage, so capacitor C_X begins to charge and voltage v_X starts to increase toward a final value of V_P .



Schmitt Trigger Oscillator

The general equation for the voltage across a capacitor in an RC network is

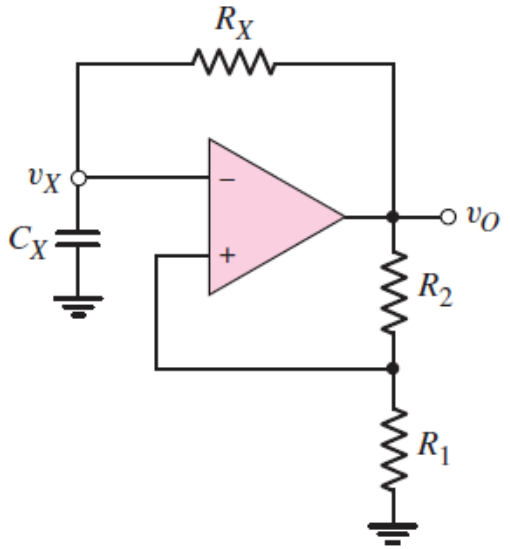
$$v_X = v_{\text{Final}} + (v_{\text{Initial}} - v_{\text{Final}})\, e^{-t/\tau}$$

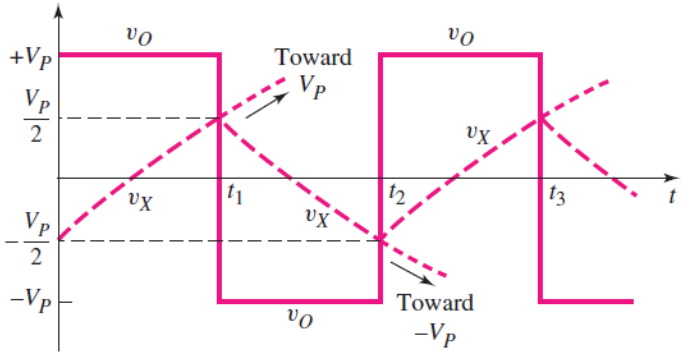
$$v_X = V_P + \left(-\frac{V_P}{2} - V_P\right) e^{-t/\tau_x}$$

$$v_X = V_P - \frac{3V_P}{2} e^{-t/\tau_x}$$

where $\tau_x = R_X C_X$

$$R_1 = R_2 \equiv R$$

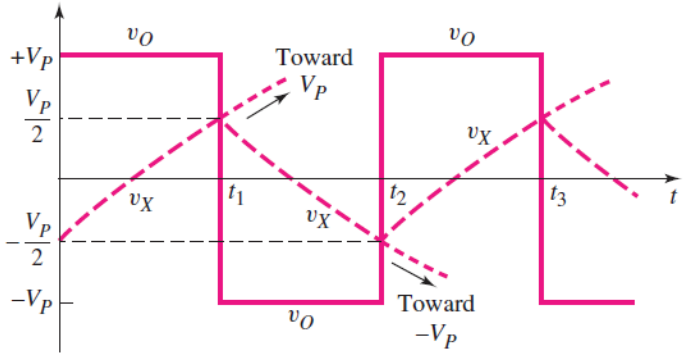




Voltage v_X increases exponentially with time toward a final voltage V_P. However, when v_X becomes just slightly greater than $v_+ = +(1/2)V_P$, the output switches to its low state of $v_O = -V_P$ and $v_+ = -(1/2)V_P$.

Schmitt Trigger Oscillator

The $R_X C_X$ network sees a negative step change in voltage, so capacitor C_X now begins to discharge and voltage v_X starts to decrease toward a final value of $-V_P$.



$$v_X = -V_P + \left[+\frac{V_P}{2} - (-V_P)\right] e^{-(t-t_1)/\tau_x}$$

$$v_X = -V_P + \frac{3V_P}{2} e^{-(t-t_1)/\tau_x}$$

Time t_1 can be found by setting $t = t_1$ when $v_X = V_P/2$,

$$\frac{V_P}{2} = V_P - \frac{3V_P}{2} e^{-t_1/\tau_x}$$

$$t_1 = \tau_x \ln 3 = 1.1 R_X C_X$$

$$T = 2.2 R_X C_X \qquad f = \frac{1}{T} = \frac{1}{2.2 R_X C_X}$$

Example Problem 15.8

Objective: Design a Schmitt trigger oscillator for a specified frequency.

Specifications: Assume that an ideal comparator is available. Use standard-valued resistors and capacitors in the final design.

Consider the oscillator in Figure 15.35. Design the circuit to oscillate at $f_o = 1$ kHz.

Solution: Using Equation (15.88), we can write

$$R_X C_X = \frac{1}{2.2 f_o} = \frac{1}{2.2(10^3)} = 4.55 \times 10^{-4}$$

If $C_X = 0.1\,\mu$F, then $R_X = 4.55\,\text{k}\Omega$.

Multivibrator

Multivibrator is an electronic circuit used to implement 2-state devices, such as oscillator, timer, flip flop, etc.

2-state: 2 voltage levels V_{high} and V_{low}
In digital electronics, Logic 1 and Logic 0

Multivibrators can be made with op-amp / 555 timer / transistor.

Based on the number of stable states, multivibrators can be categorized into 3 types

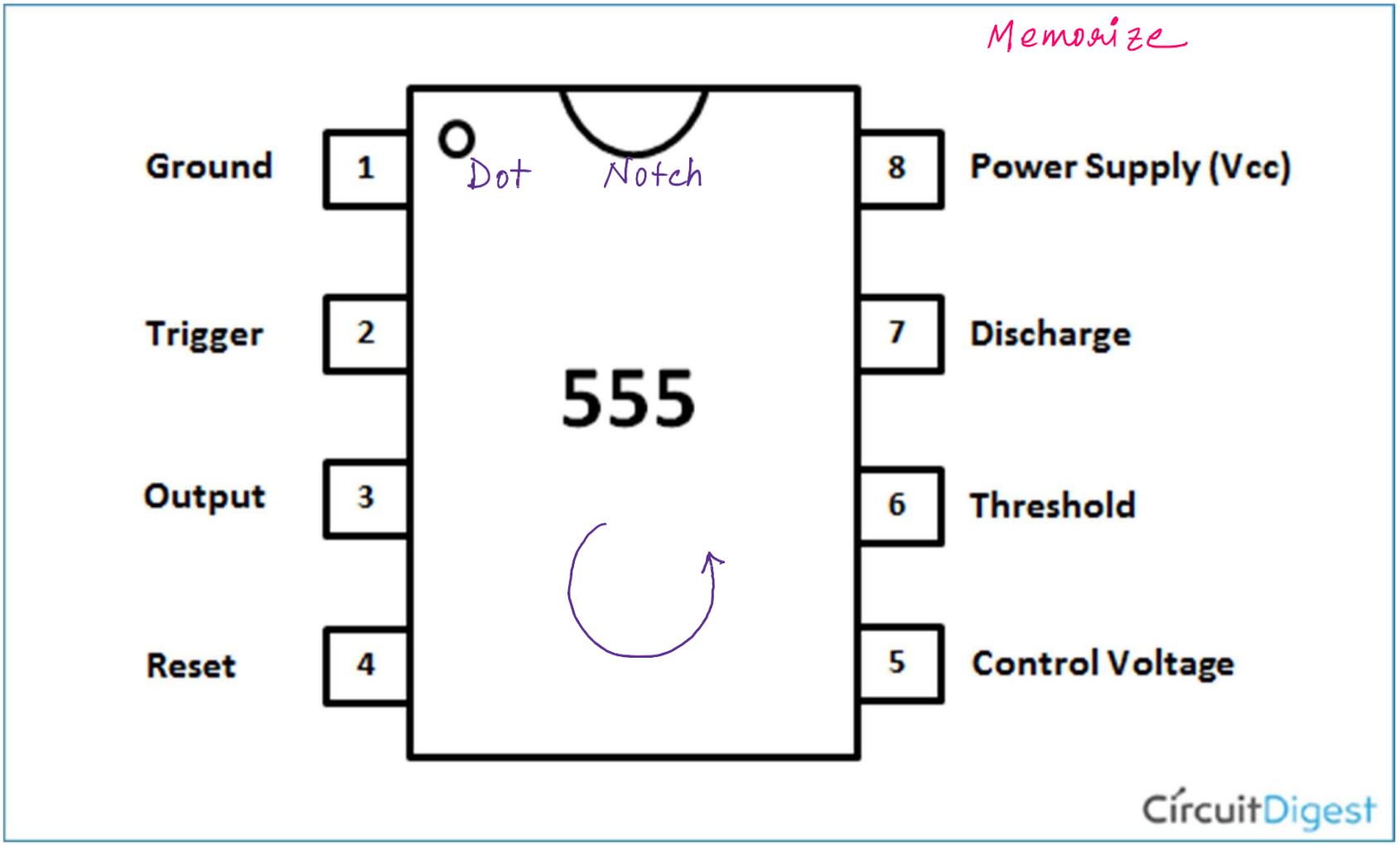
555 Timer

Popular in timing related applications

- ✓ Tone and alarm generation
- ✓ Frequency division
- ✓ Provide timing delays
- ✓ As relaxation oscillator

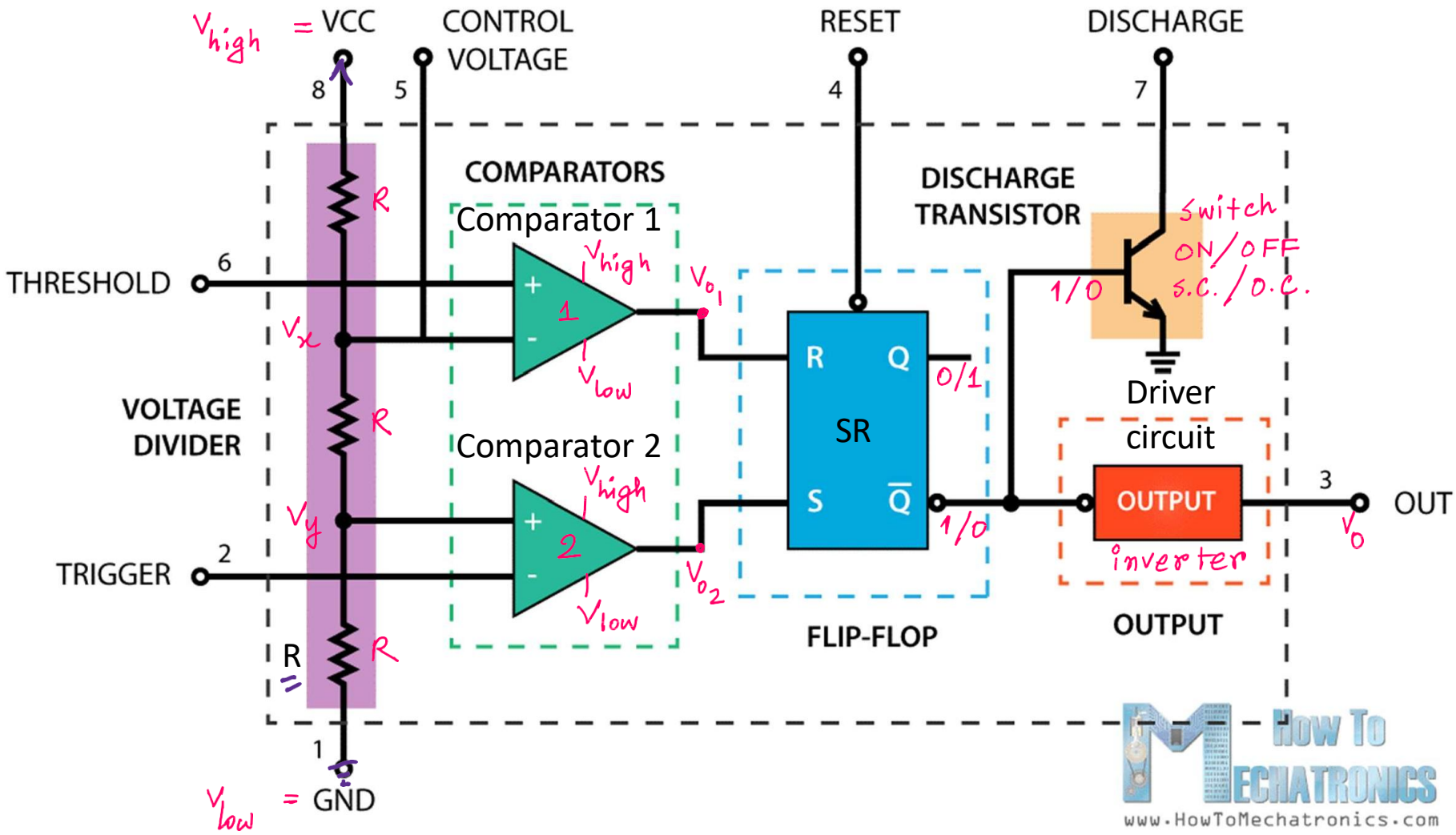
555 Timer – Pin diagram

Dual-in line packaging



8 pin DIP Package

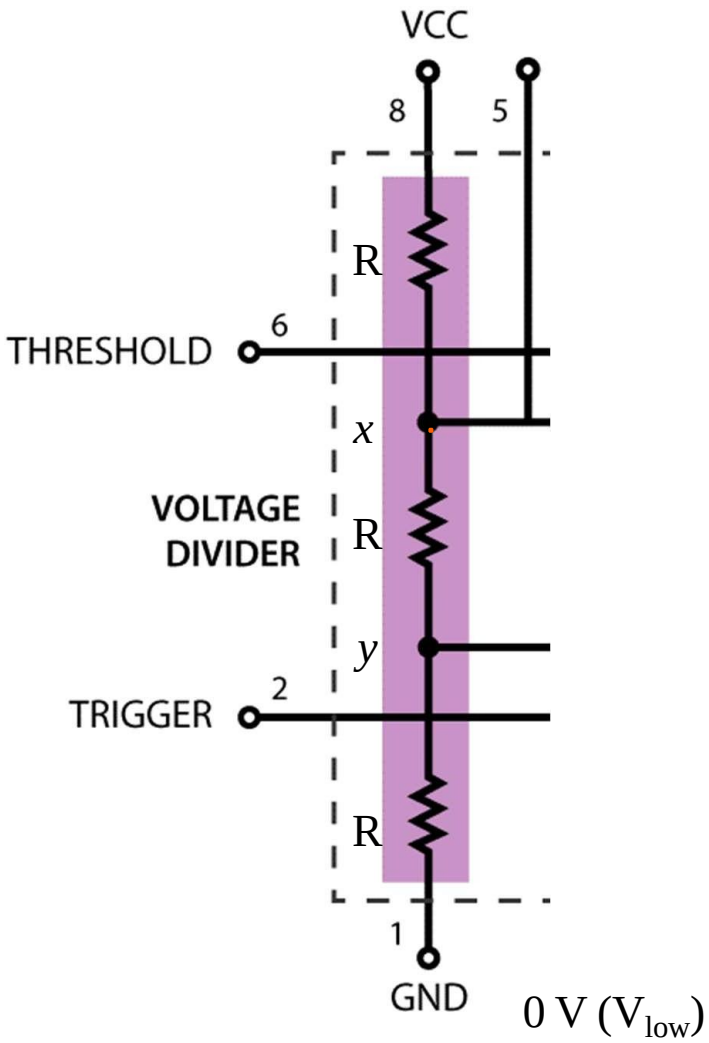
555 Timer – Block diagram



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555 Timer – Voltage divider network

Supply voltage pin: 4.5 to 15 V (V_{high})



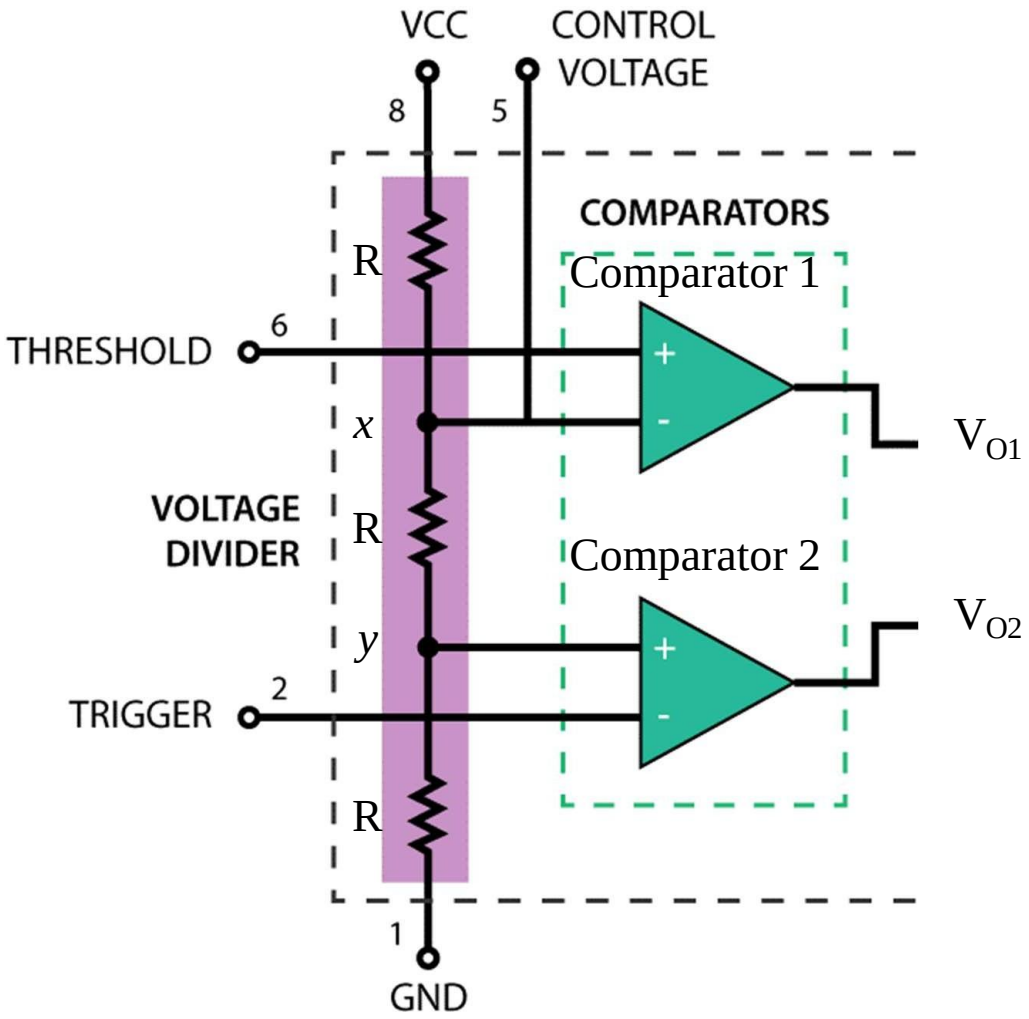
Applying voltage divider rule:

$$V_x = \frac{2R}{3R} V_{CC} = \frac{2}{3} V_{CC}$$

$$V_y = \frac{R}{3R} V_{CC} = \frac{1}{3} V_{CC}$$

555 Timer – comparators

Pin 6 and pin 2 can change the timings of the output signal



Pin 6, V_x: input of comparator 1

$V_{threshold} > V_x$: $V_{O1} = V_{high}$

$V_+ > V_-$

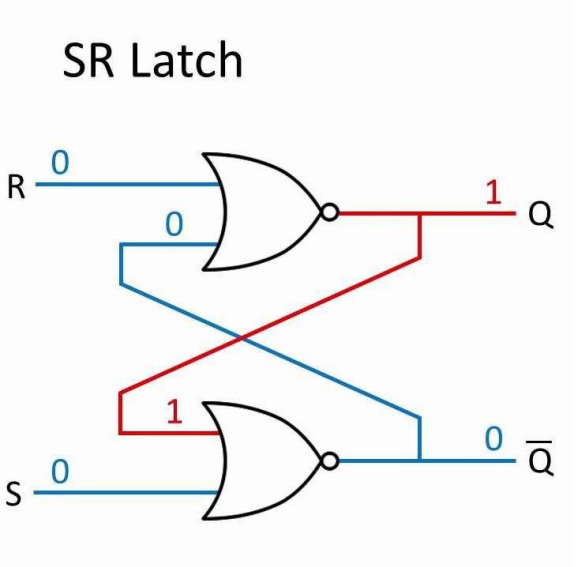
Pin 2, V_y: input of comparator 2

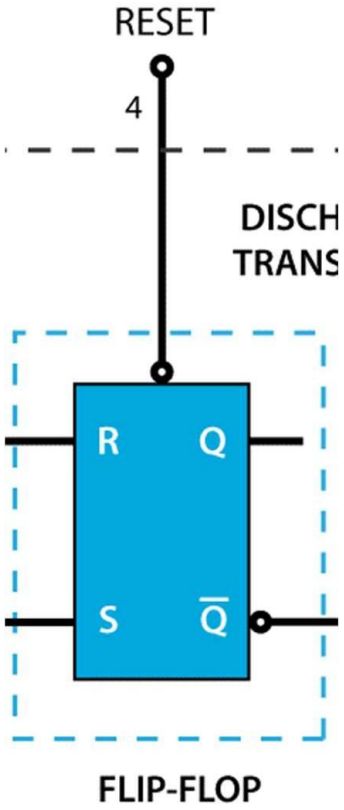
$V_{trigger} < V_y$: $V_{O2} = V_{high}$

$V_- < V_+$

SR Flip-flop

S	R	Q(n)	Q(n+1)
0	0	0/1	0/1
0	1	0/1	0 (RESET)
1	0	0/1	1 (SET)
1	1	0/1	× (INVALID)



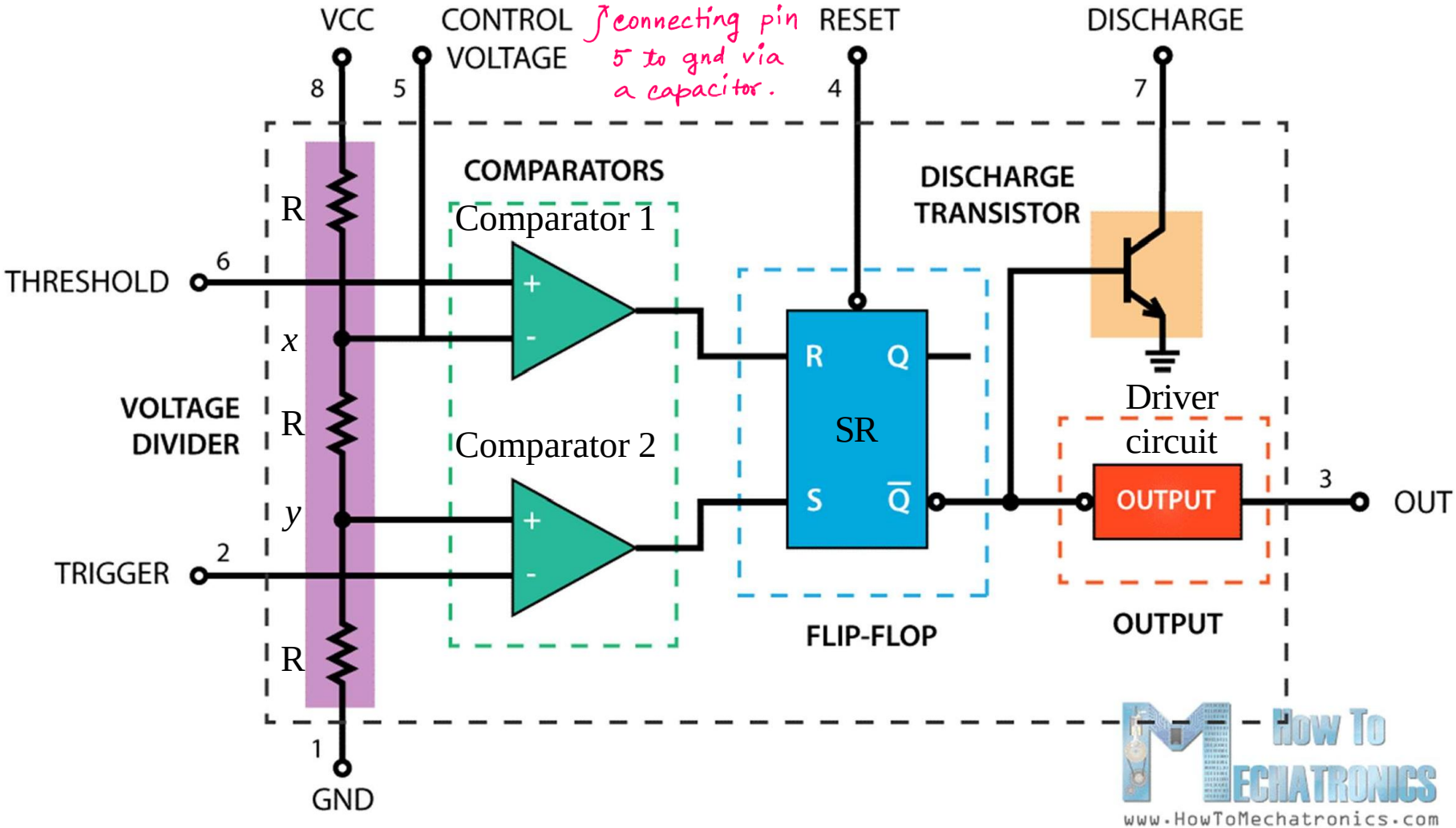


555 Timer – Block diagram

External R and C are connected between pin 6 (threshold) and 7(discharge) between pins 8 (Vcc) and 1 (Gnd)

Externally other V_{ref} can be applied to comparator 1, it can change the o/p signal timing, too

We will deactivate control voltage option by connecting pin 5 to gnd via a capacitor.

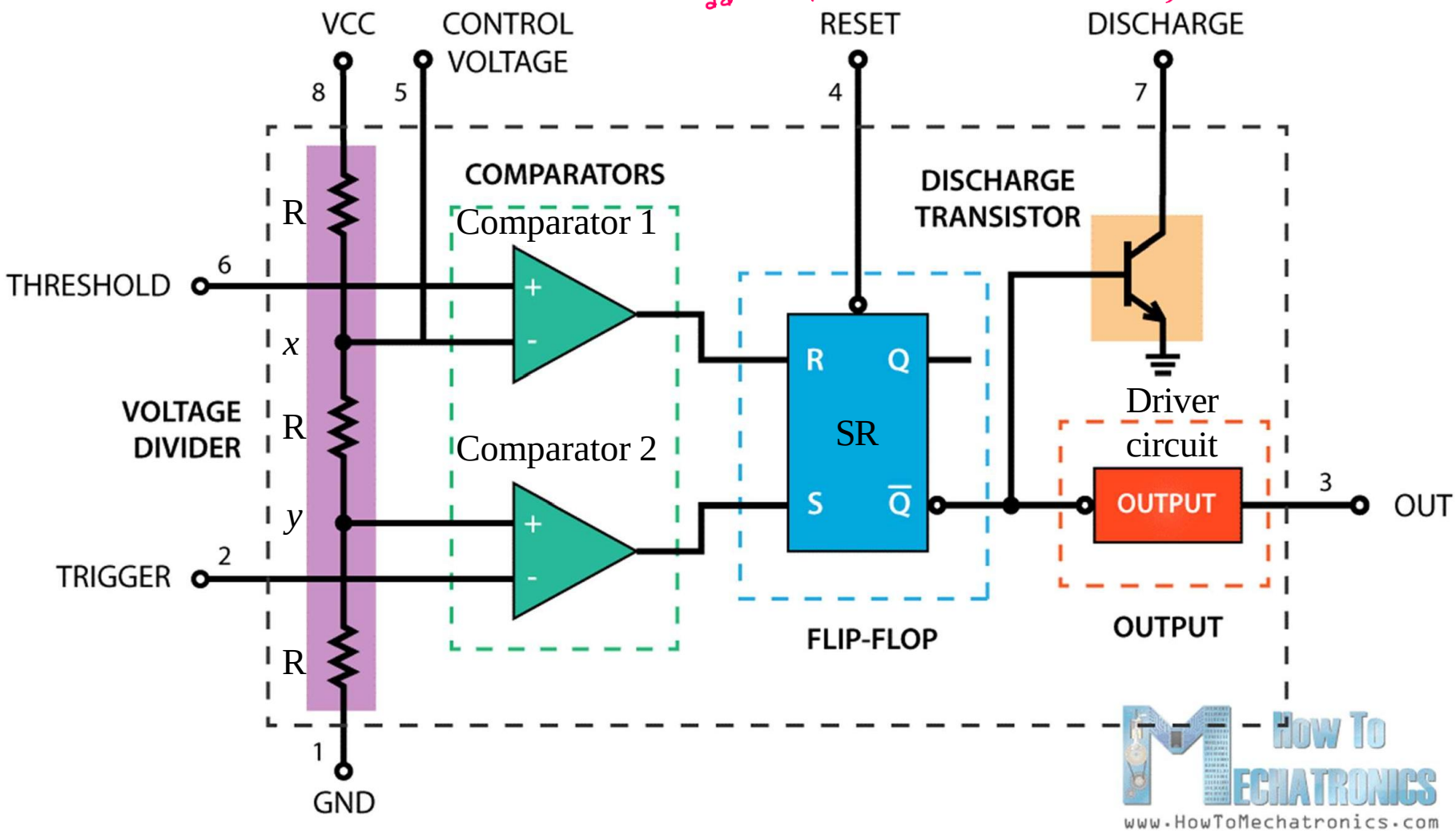


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555 Timer – Block diagram

Reset pin: Logic 0 (Active low pin)
It resets the flip-flop and o/p of flip-flop

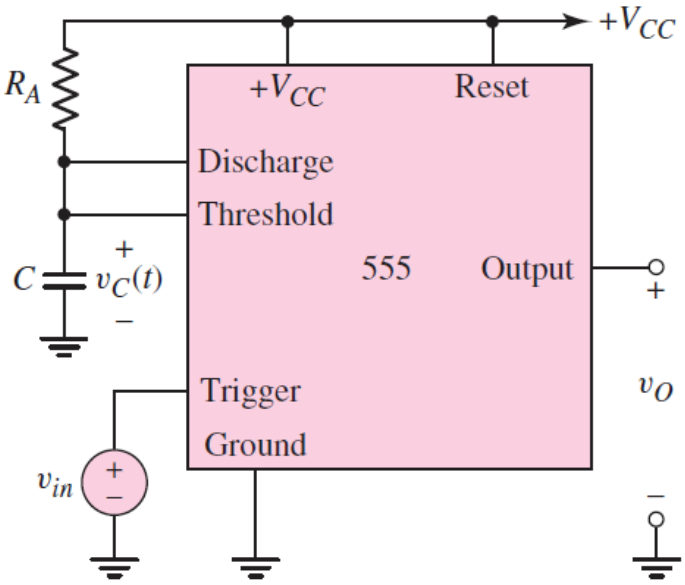
We will deactivate reset pin by applying V_{dd} at pin 4 (∵ active low pin).



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Monostable Multivibrator

A monostable multivibrator, also called a one-shot, operates by charging a timing capacitor with a current set by an external resistance.

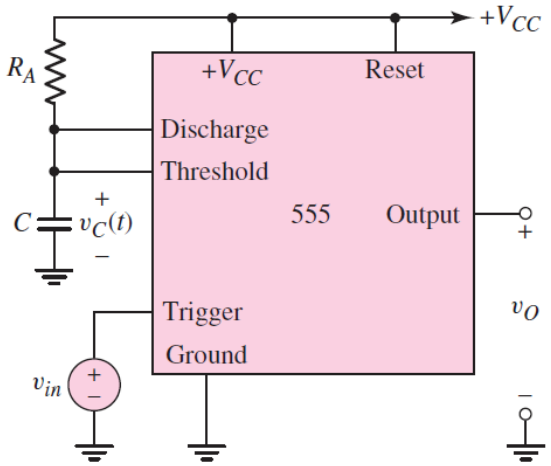


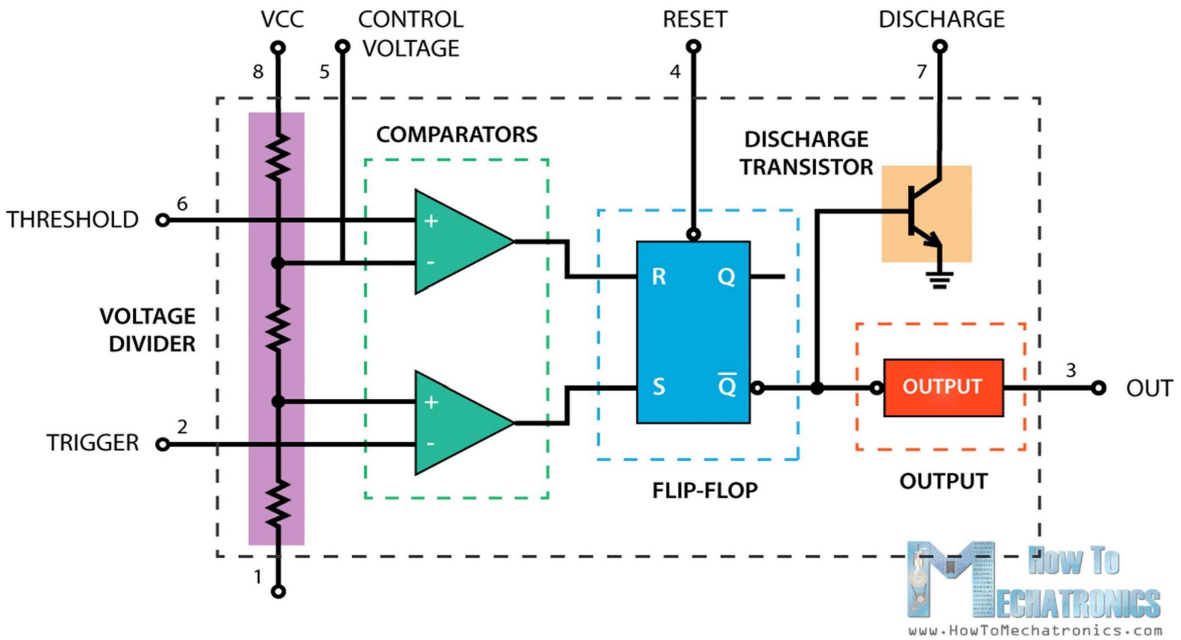
The 555 circuit connected as a monostable multivibrator

With a high voltage V_+ applied to the trigger input, the trigger comparator output is low, the flip-flop output .Q is high.

The discharge transistor is turned on, and the timing capacitor C is discharged to nearly ground potential. The output of the 555 circuit is then low,

Monostable Multivibrator

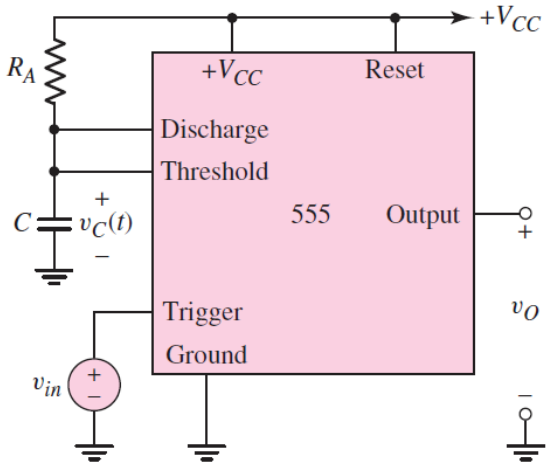


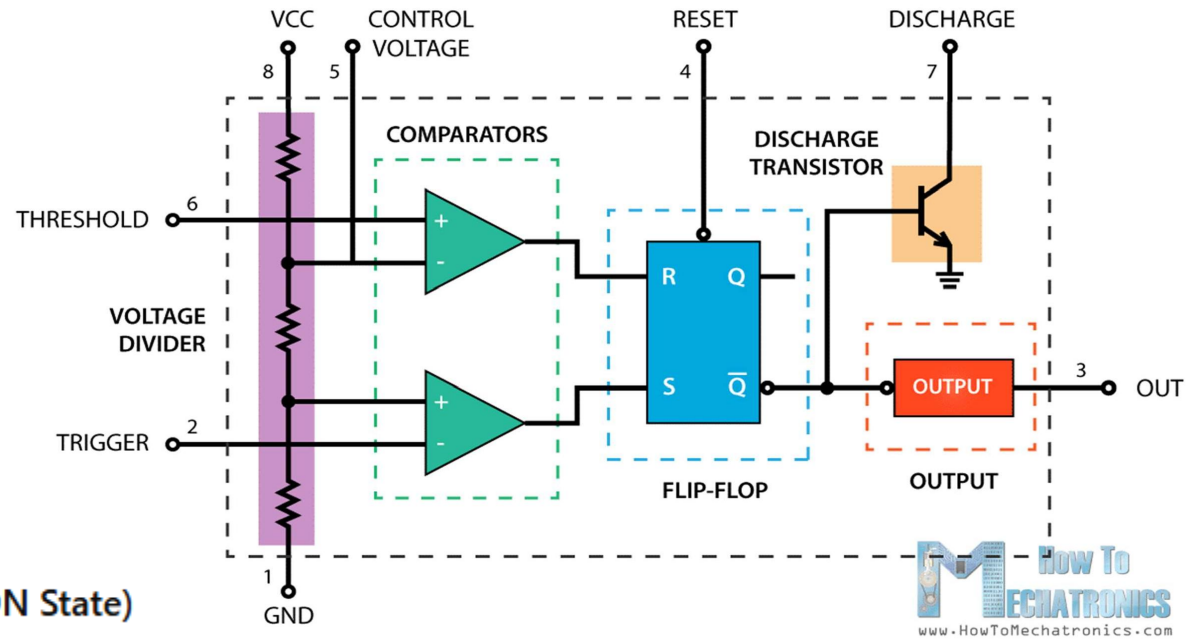


Step 1: Negative Trigger Signal

- A **negative pulse** (voltage drops below **1/3 Vcc**) is applied to the **Trigger input (pin 2)**.
- The **Trigger Comparator** detects this and **sets the SR Flip-Flop**.
- This makes the **output HIGH (Q = 1)**.
- At the same time, the **Discharge Transistor turns OFF**, allowing the capacitor to **start charging** through resistor R.

Monostable Multivibrator





Step 2: Capacitor Charging (Unstable ON State)

- The capacitor starts charging from 0V towards Vcc via the resistor.
- Voltage across the capacitor $V_C(t) = V_{cc}(1 - e^{-t/RC})$.
- As long as $V_C < 2/3Vcc$, nothing changes.

Step 3: Threshold Reached (Return to Stable State)

- Once V_C reaches **2/3 Vcc**, the **Threshold Comparator** triggers.
- It **resets the SR Flip-Flop**.
- The **output goes LOW** (Q = 0).
- The **Discharge Transistor turns ON**, discharging the capacitor quickly to 0V.

Back to Initial State Now the system is **back to the stable state** (OFF), waiting for another trigger.

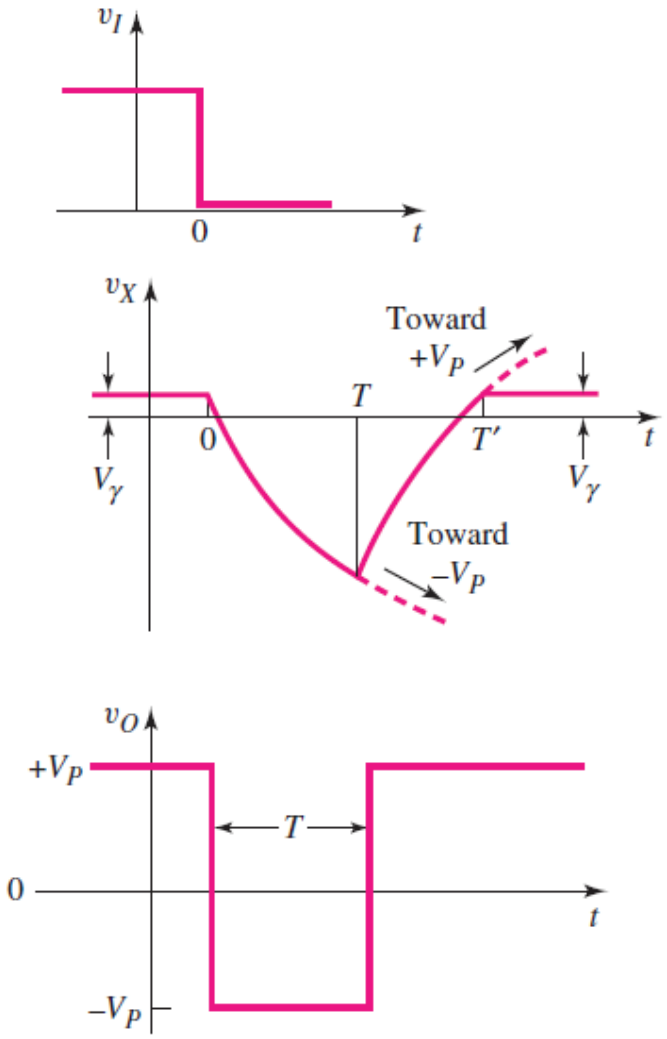
Monostable Multivibrator

When a negative-going pulse is applied to the trigger input, the output of the trigger comparator goes high when the trigger pulse drops below *(1/3)*V_+.
Then output Q goes low, which means that the output of the 555 goes high, and the discharge transistor turns off.

The output of the 555 remains high even if the trigger pulse returns
to its initial high value, because the reset input to the flip-flop is still low.

The capacitor voltage is given by

$$v(t) = V^+(1 - e^{-t/RC})$$



Monostable Multivibrator

The width of the output pulse is determined by setting v(t) = (2/3)V+ and t = T

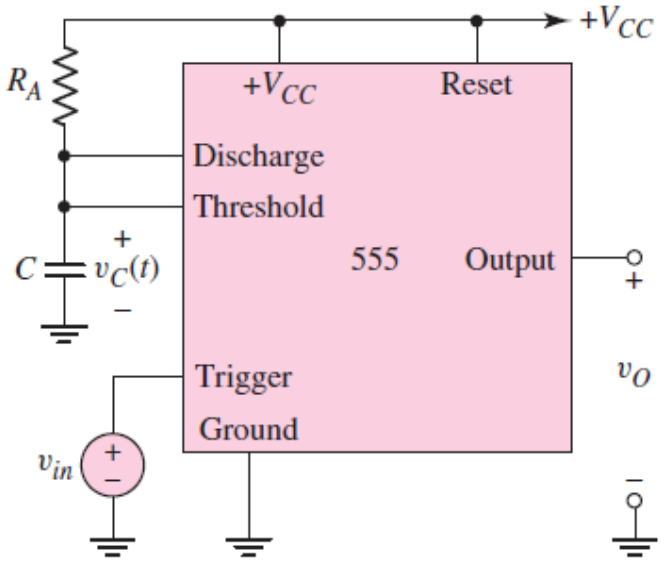
$$\left(\frac{2}{3}\right)V^{+} = V^{+}\left(1 - e^{-T/RC}\right)$$

$$T = RC\,\ln(3) = 1.1\,RC$$

Example Problem 15.10

Objective: Design a 555 IC as a monostable multivibrator to produce a specified output pulse width.

Specifications: The circuit with the configuration shown in Figure 15.40 is to be designed to produce an output pulse width of 100 μs.



Solution:

Let $C = 15\,\text{nF}$.

$$R = \frac{T}{1.1C} = \frac{100 \times 10^{-6}}{(1.1)(15 \times 10^{-9})} \Rightarrow 6.06\,\text{k}\Omega$$

Astable Multivibrator (Free running) - Review

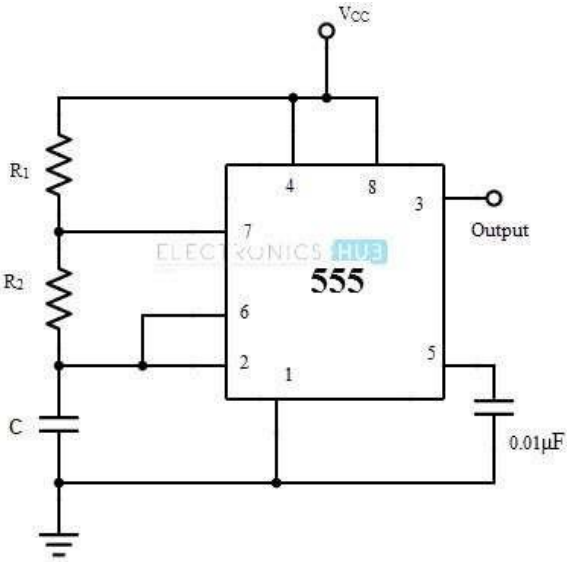
Output is not stable in any of the two states.

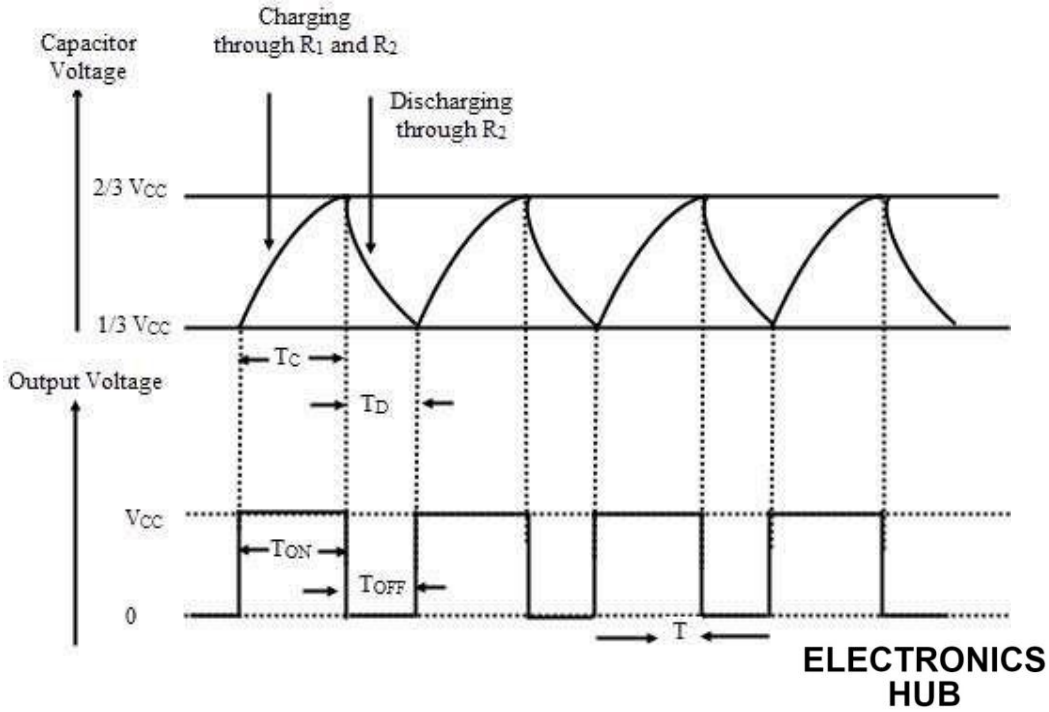
Output continuously change between the 2 states.

There is no input trigger

T_{high} and T_{low} depend on externally connected R and C

Application: relaxation oscillator





ELECTRONICS HUB

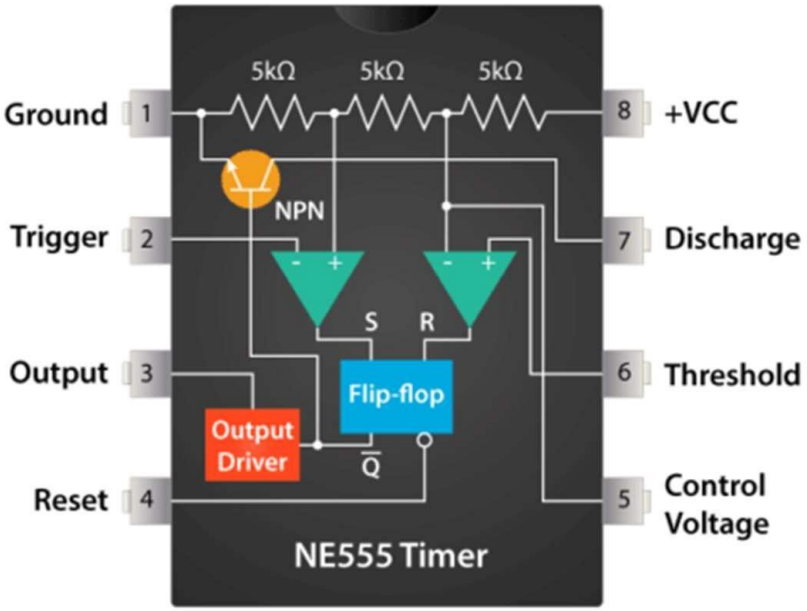
Astable Multivibrator

Reset (4) and control (5) pins are not used.

Reset pin (4) is connected to $V_{high} = V_{cc}$ (∵ active low).

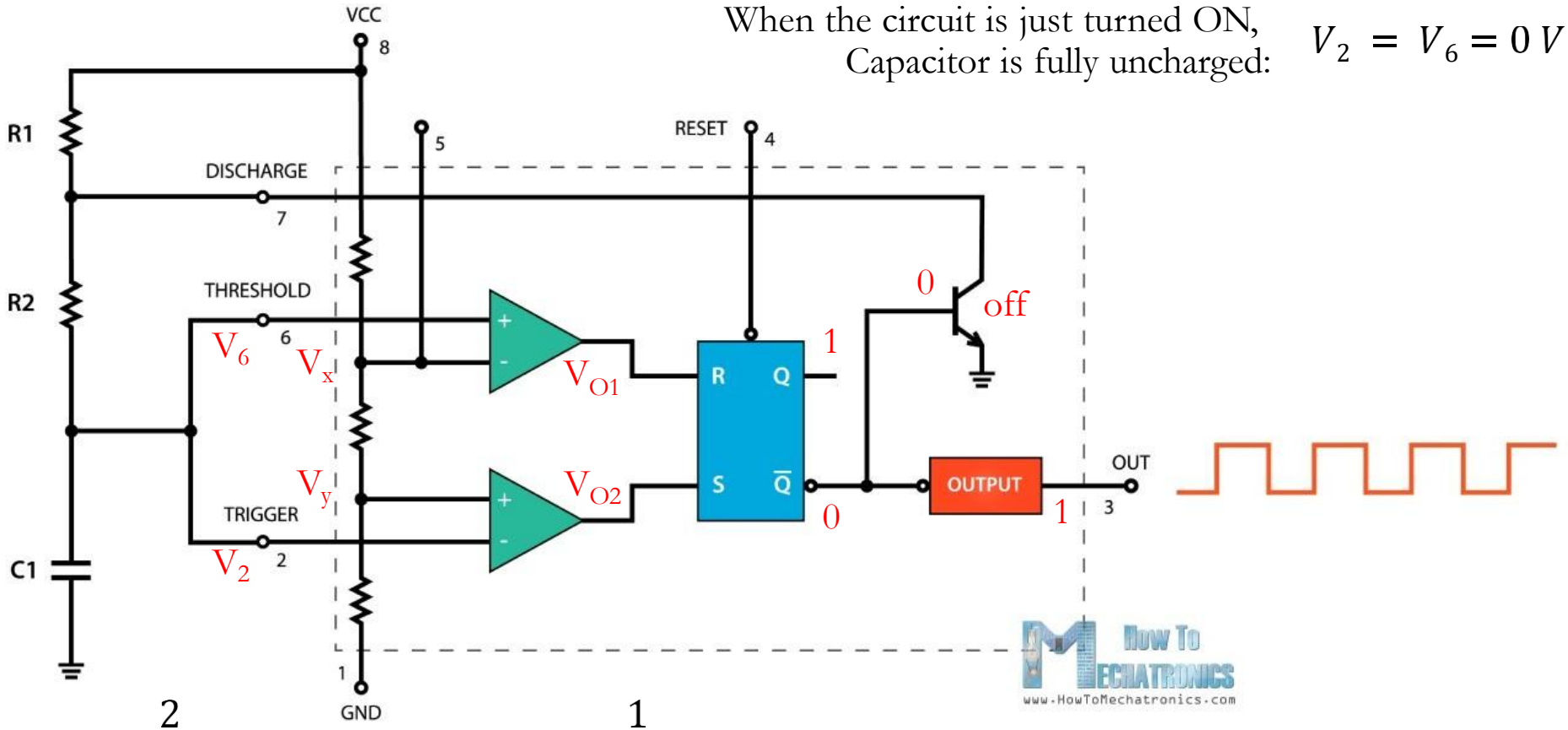
Control pin (5) is connected to $V_{low} = 0$ V via a capacitor.

Threshold (6) and trigger (2) pins are connected (∵ no input trigger).



Astable Multivibrator

When the circuit is just turned ON,
Capacitor is fully uncharged: $V_2 = V_6 = 0\ V$



$V_x = \frac{2}{3}V_{CC}$ $\qquad$ $V_y = \frac{1}{3}V_{CC}$

$V_x > V_2$ $\qquad$ $V_y > V_6$

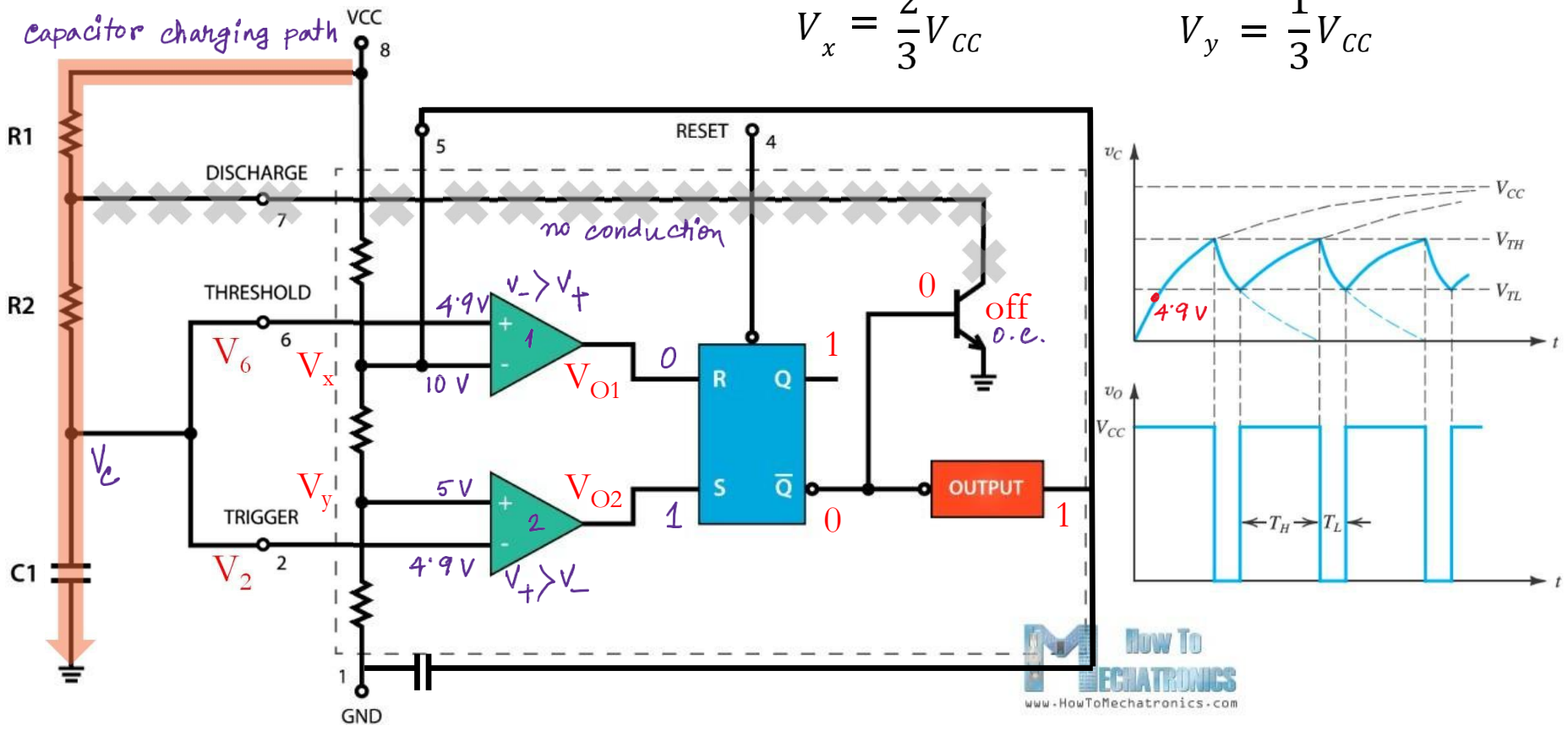
$V_{O1} = V_{low}$ $\qquad$ $V_{O2} = V_{high}$

$R = 0$ $\qquad$ $S = 1$

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Astable Multivibrator

$$V_x = \frac{2}{3}V_{CC} \qquad V_y = \frac{1}{3}V_{CC}$$



When $Q = 1: Q' = 0$

Transistor is OFF

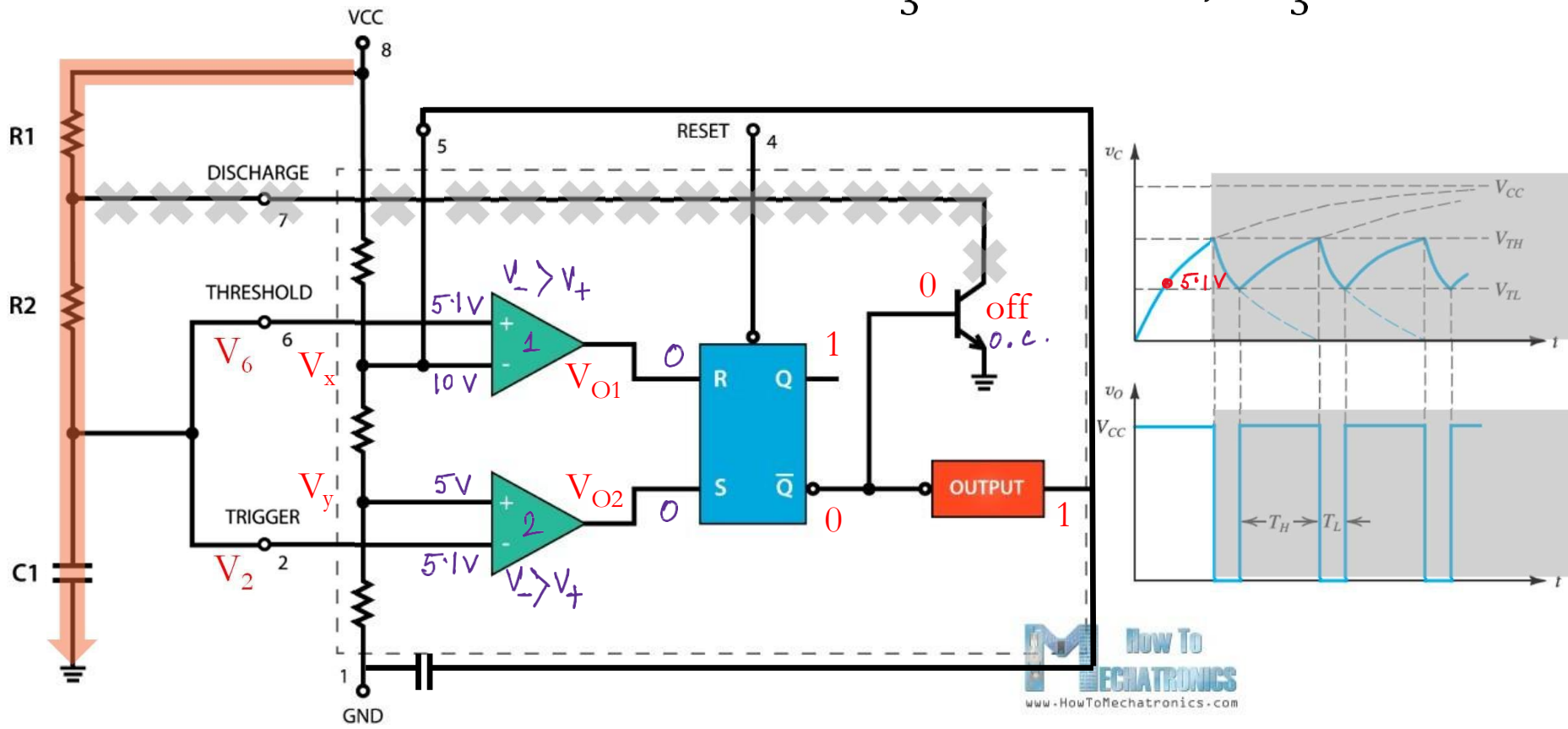
$V_o = V_{high}$

Capacitor will charge through R_1 and R_2

$V_2 = V_6$ will gradually increase

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_2 > Vy$:

$V_{O2} = V_{low}$

$R = 0$ $S = 0$ **Memory state**

C_1 will continue charging as long as $V_c < V_x$.

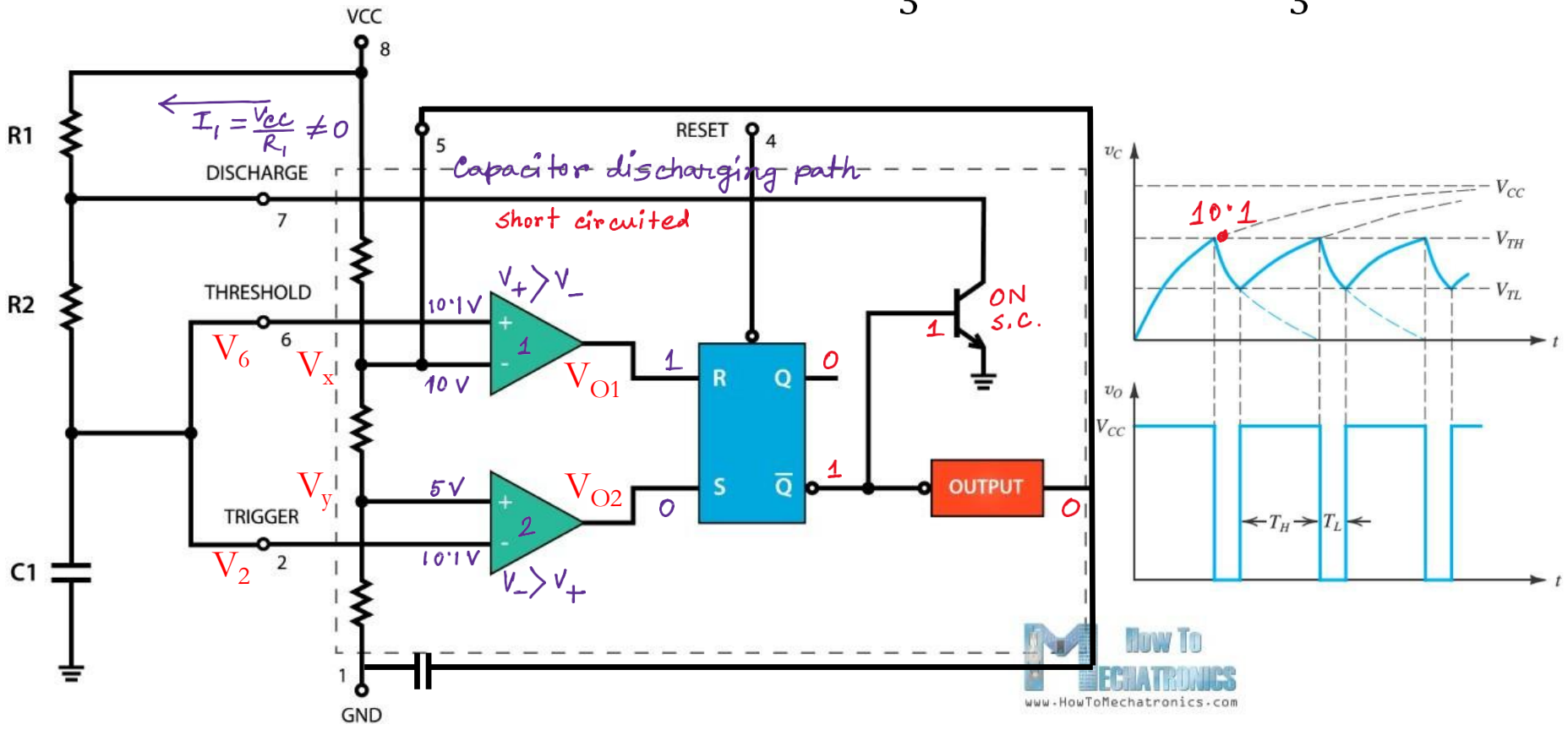
$Q = 1 : Q' = 0$

Transistor is OFF

$V_o = V_{high}$

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_6 > Vx$:

$V_{O1} = V_{high}$

∴ As soon as $V_c > V_x$, C_1 interrupts charging and instead starts discharging.

$R = 1$ $S = 0$ **Reset state** $Q = 0: Q' = 1$

Capacitor will discharge through R_2

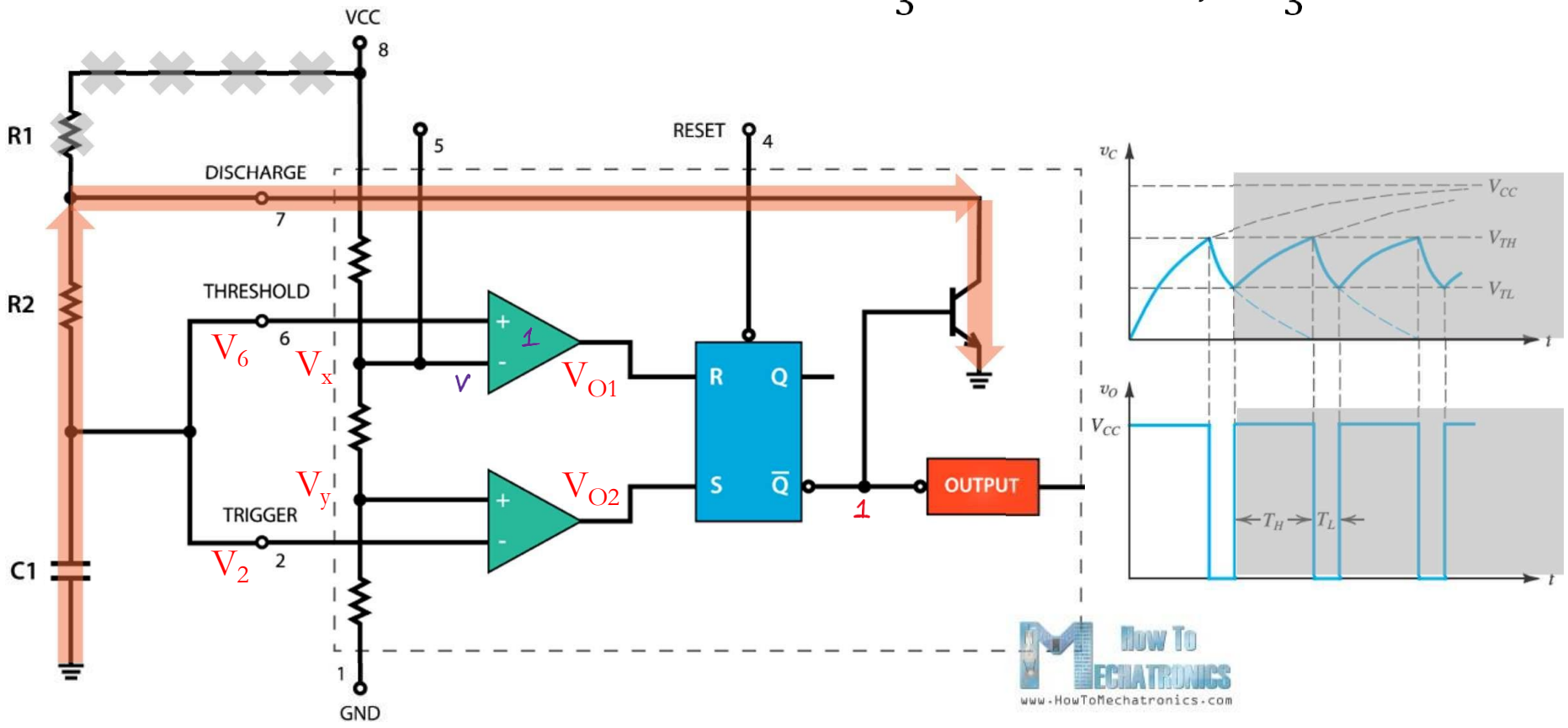
$V^2 = V^6$ will gradually decrease

Transistor is ON

$V_o = V_{low}$

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_6 < Vx$:

$V_{O1} = V_{low}$

$R = 0$ $S = 0$ **Memory state** $Q = 0: Q' = 1$

Transistor is ON

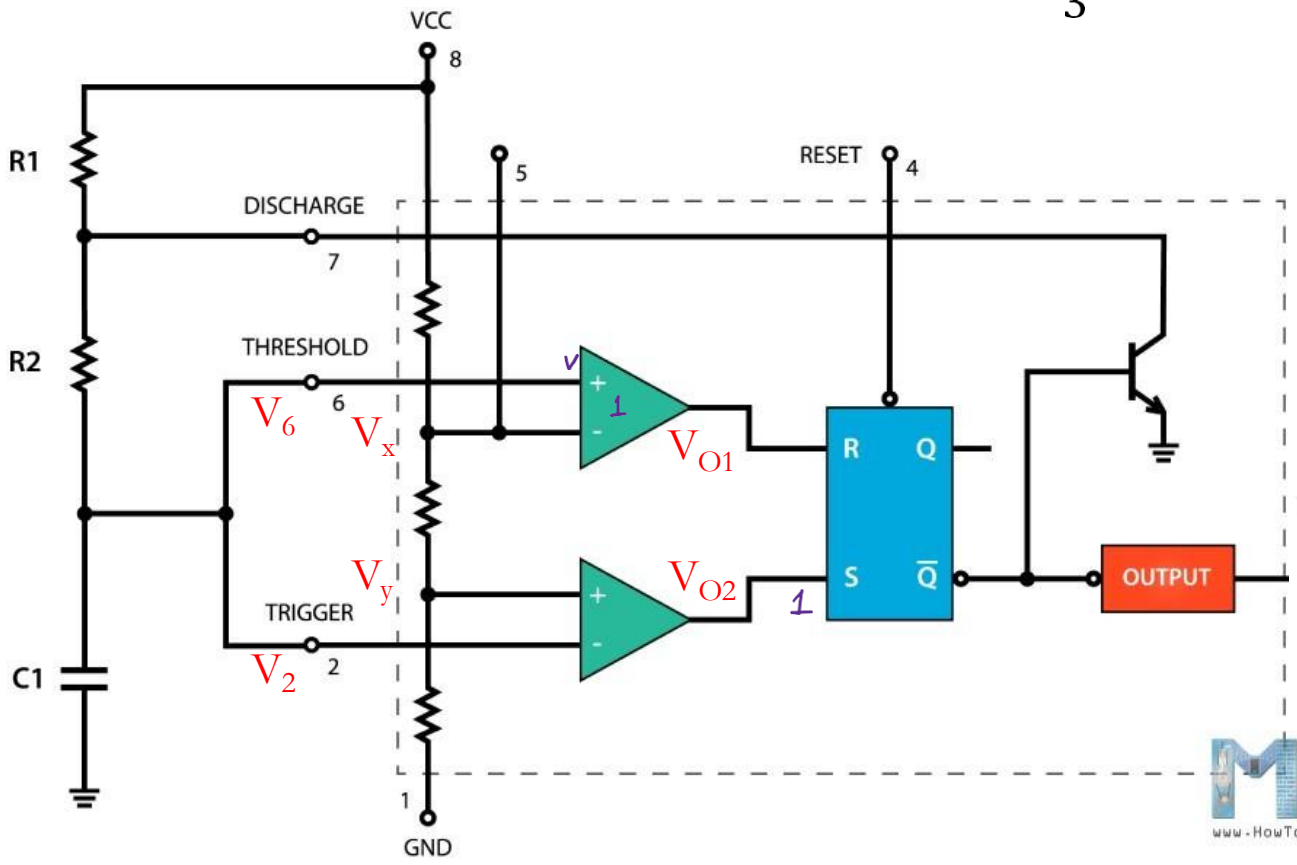
Capacitor will discharge through R_2

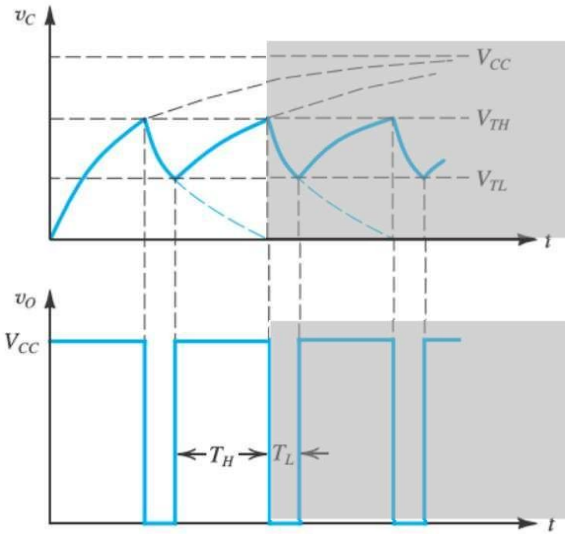
$V_2 = V_6$ will gradually decrease

$V_o = V_{low}$

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$







When $V_2 < Vy$:

$V_{O2} = V_{high}$

$R = 0$ $S = 1$ **Set state**

$Q = 1: Q' = 0$

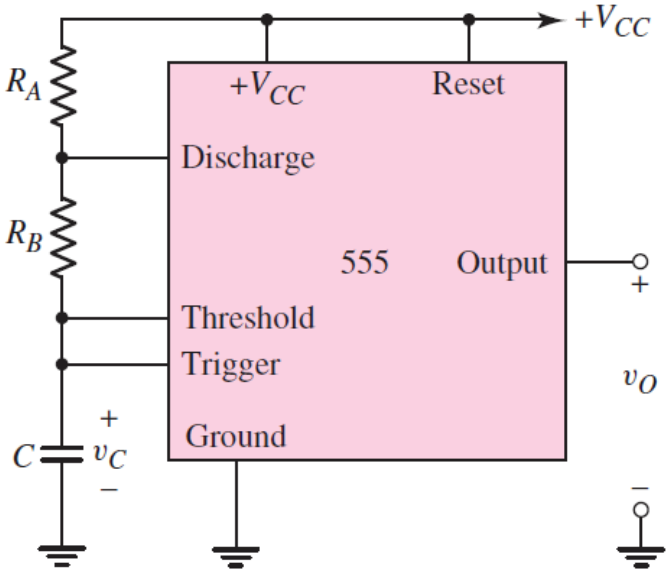
Transistor is OFF

$V_o = V_{high}$

Capacitor will charge through R_1 and R_2

$V_2 = V_6$ will gradually increase

Astable Multivibrator



Astable multivibrator 555 circuit

In the astable mode, the timing capacitor C charges through $R_A = R_B$ until $v(t)$ reaches $(2/3)V_+$. The threshold comparator output then goes high, forcing the flip-flop output Q to go high

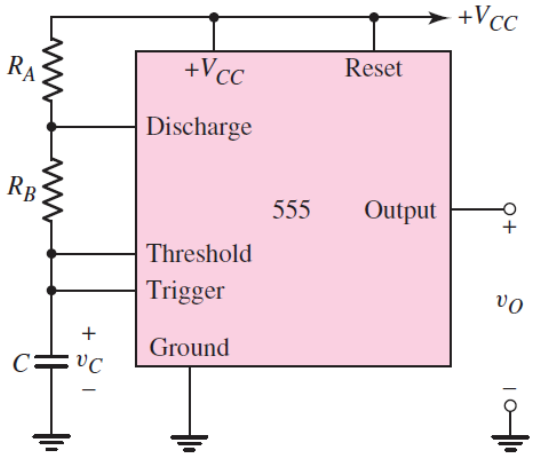
The discharge transistor turns on, and the timing capacitor C discharges through R_B and the discharge transistor. The capacitor voltage decreases until it reaches (1/3)V+, at which point the trigger comparator switches states and sends Q low.

When v(t) reaches the threshold level of (2/3)V+, the cycle repeats itself.

Astable Multivibrator

When the timing capacitor is charging, during the time 0 < t < TC

$$v(t) = \frac{1}{3}V^+ + \frac{2}{3}V^+(1 - e^{-t/\tau_A})$$ where $\tau_A = (R_A + R_B)C$.



At time $t = T_C$ the capacitor voltage reaches the threshold level

$$v(T_C) = \frac{2}{3}V^+ = \frac{1}{3}V^+ + \frac{2}{3}V^+(1 - e^{-T_C/\tau_A})$$

$$T_C = \tau_A \ln(2) = 0.693(R_A + R_B)C$$

When the timing capacitor is discharging, during the time $0 < t < T_D$

$$v(t') = \frac{2}{3}V^+ e^{-t'/\tau_B}$$ where $\tau_B = R_B C$.

Astable Multivibrator

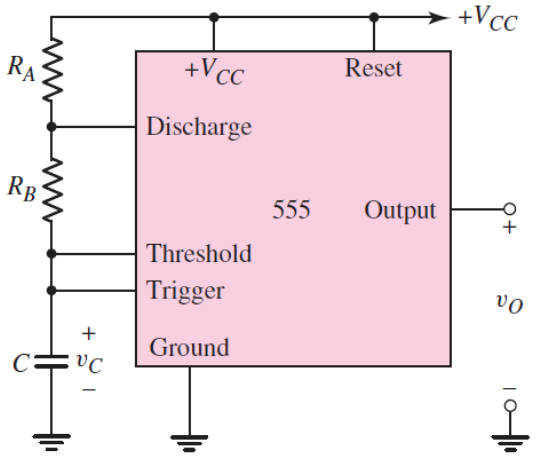
At time $t = T_D$, the capacitor voltage reaches the trigger level

$$v(T_D) = \frac{1}{3}V^+ = \frac{2}{3}V^+ e^{-T_D/\tau_B}$$

$$T_D = \tau_B \ln(2) = 0.693 R_B C$$

$$f = \frac{1}{T} = \frac{1}{T_C + T_D} = \frac{1}{0.693(R_A + 2R_B)C}$$

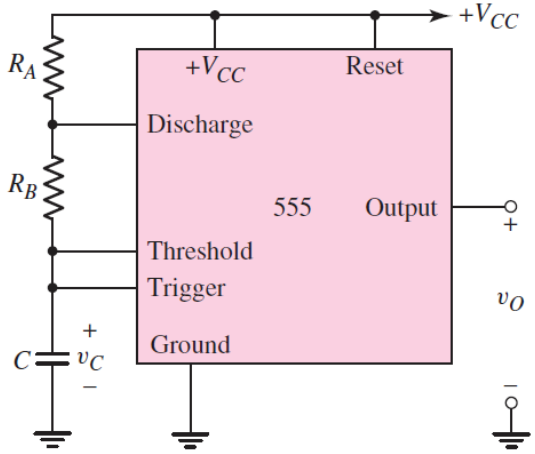
$$\text{Duty cycle} = \frac{T_C}{T} \times 100\% = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$



Example Problem 15.8

Objective: Design the 555 IC as an astable multivibrator for a specified frequency and duty cycle.

Specifications: The circuit with the configuration in Figure 15.41 is to be designed such that the frequency is 50 kHz and the duty cycle is 75 percent.



Solution: The frequency of oscillation,

$$f = \frac{1}{0.693(R_A + 2R_B)C}$$

Therefore,

$$R_A + 2R_B = \frac{1}{(0.693)fC} = \frac{1}{(0.693)(50 \times 10^3)(1 \times 10^{-9})} \Rightarrow 28.9\,\text{k}\Omega$$

$$\text{Duty cycle} = 0.75 = \frac{R_A + R_B}{R_A + 2R_B}$$

Example Problem 15.11

$$0.75 = \frac{R_A + R_B}{R_A + 2R_B}$$

which yields

$$R_A = 2R_B$$

Combining Equations of R_A and R_B

$$R_A = 14.5\,\text{k}\Omega \qquad \text{and} \qquad R_B = 7.23\,\text{k}\Omega$$